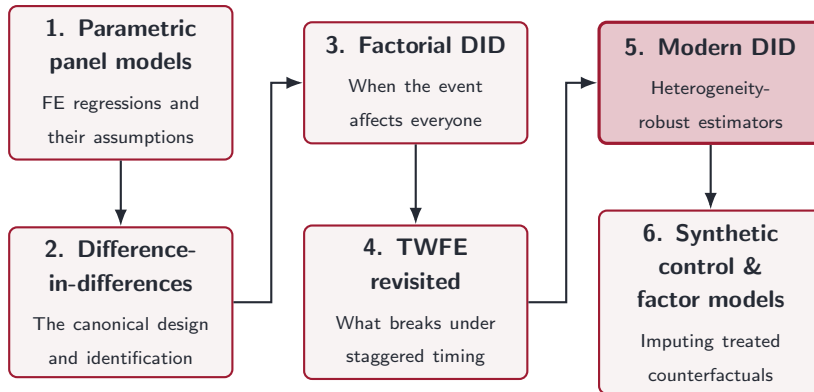


Panel Methods (5): Modern DID

Lectures on Causal Panel Analysis

Yiqing Xu
Stanford University

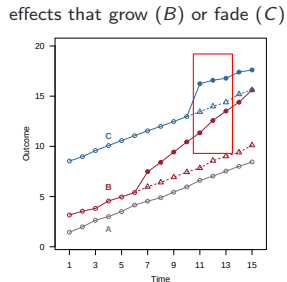
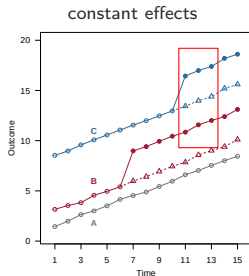
September 2026



- Lecture 4 diagnosed the problem: TWFE on staggered panels mixes clean and contaminated comparisons, and parallel trends failures are common
- This lecture is about the remedies:
 - A tour of the **HTE-robust estimators**: what each one compares, and when each applies
 - The **diagnostics** that come with the imputation approach
 - Three worked examples, and what we learned from **reanalyzing 49 published studies** with these tools
- Lecture 6 introduces low-rank methods, synthetic control and factor models, that can address some parallel trends violations

Review: What Goes Wrong under Heterogeneous Effects

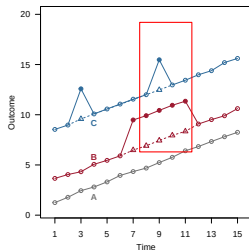
- Three units: A never treated, B treated from period 7, C from period 11; solid = observed, dashed = the untreated trajectory $Y(0)$
- **Result** (Goodman-Bacon 2021 and others): with heterogeneous effects, TWFE need not recover any convex combination of the individual effects, even when parallel trends holds
- **Intuition**: in the boxed periods, B 's treated observations serve as the comparison for C , the “forbidden comparison”; harmless while B 's effect is constant, biased once it keeps growing
- The estimators in this lecture avoid such comparisons by construction



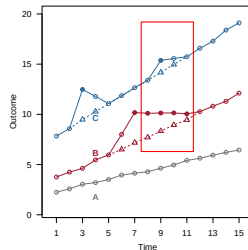
...and It Compounds with Reversal and Carryover

- Most political science panels have **treatment reversal**: here B is treated in periods 7–11 and then exits, and C switches on twice
- In the boxed window, B and C serve as comparisons for each other while both are treated
- Effects that build up and **linger after exit** (carryover) make it worse: the post-exit periods are no longer clean comparisons either
- Add parallel trends violations and anticipation, and the single-coefficient TWFE regression stands little chance; we need estimators that say **which cells compare with which**

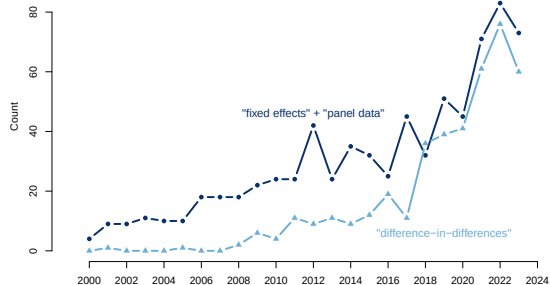
effects switch off with the treatment



effects build up and linger after exit



The Menu Has Grown



- In our review of top political science journals, 77% of panel papers use FE models, mostly TWFE
- The econometric critiques produced many new estimators in a short span
- This lecture: a map of that menu, organized by which cells each estimator compares

Preview: What the Reanalyses Will Show

Common practice

- Fixed effects models in 77% of panel papers, TWFE in 58%
- Cluster-robust SEs in 98% of the reanalyzed studies; a cluster bootstrap in 16%
- Some plot of the raw data in 65%



Do the results hold up? Yes and no

- Yes: HTE-robust estimators **rarely** flip signs
- No: parallel trends violations are still common
- No: power runs out once the estimator is HTE-robust
- No: few findings survive a **mild** sensitivity analysis

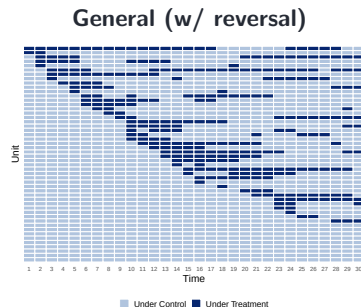
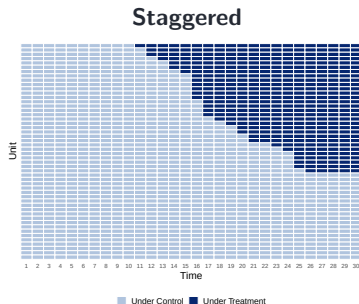
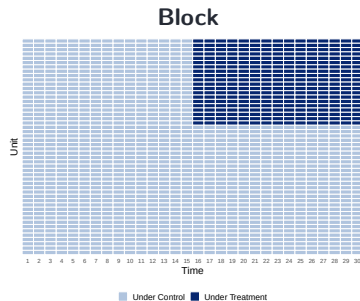
Strong empirical support for fewer than a third of the findings



Takeaways

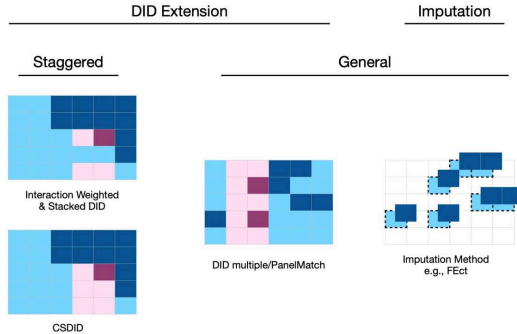
- Parallel trends, and the research design behind it, is the first-order issue
- The concern over heterogeneous effects is valid but second-order
- Validation is the key: an event-study plot is the minimum; sensitivity analysis helps
- “Robust” DID needs **a strong design** and **a lot of power**

Three Data Settings



- Which estimators are available depends on the setting; treatment reversal (present in the majority of political science studies) rules out the staggered-only tools

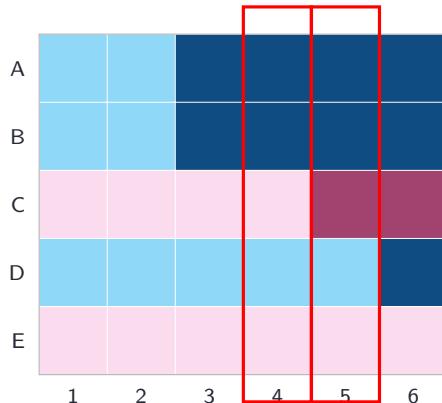
The Organizing Principle



- **DID extensions** pick explicit 2×2 comparisons; **imputation methods** fit an outcome model on untreated cells only

Sun and Abraham (2021): Interaction Weighted

- Comparison group: **never-treated**
- Estimate a cohort ATT (CATT) by 2×2 DID for each cohort g and period since treatment l
- $ATT = \text{average CATT, weighted by cohort size}$
- The same construction for every cohort; shown here: cohort C at $l = 0$, the red window pairing the reference period with the first treated period



- The building block behind these estimators is the **group-time ATT**:

$$ATT(g, t) = \mathbb{E}[Y_{it}(1) - Y_{it}(0) \mid G_i = g], \quad t \geq g$$

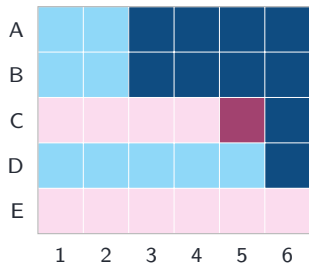
the effect for the cohort first treated at g , in calendar period t

- One number per cohort-period pair is too much detail; the art is in the **aggregation**:
 - By relative time $l = t - g$: the **event-study** plot
 - By calendar time t : does the effect grow or fade over the sample period?
 - Overall: a weighted average across all treated cohort-periods
- Different papers need different aggregations; report the one your theory speaks to, not just the overall number

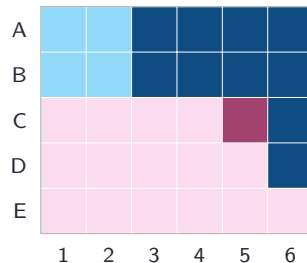
Callaway and Sant'Anna (2021)

- Comparison group: **not-yet-treated** (in addition to never-treated)
- “Doubly robust” with covariates: combines outcome regression and propensity-score weighting

Sun and Abraham (2021)



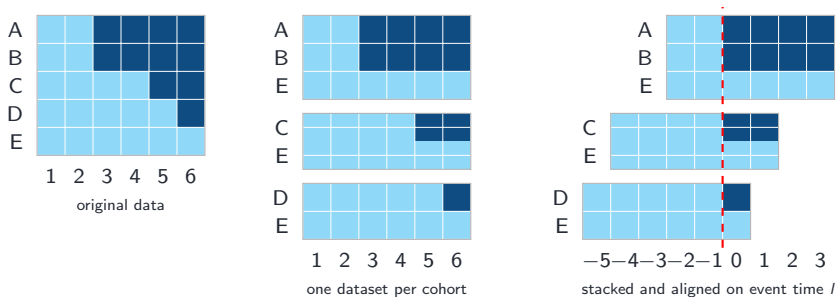
Callaway and Sant'Anna (2021)



- More comparison cells $\Rightarrow$ more power, at the price of assuming parallel trends also holds for the not-yet-treated

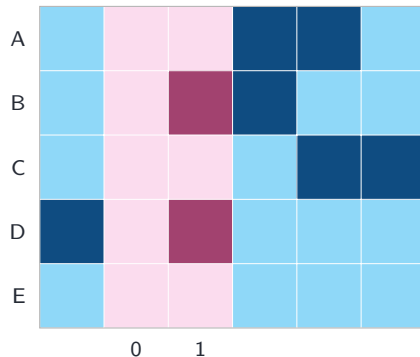
Stacked DID (Cengiz et al. 2019)

- Build one dataset per cohort: the cohort's units plus a copy of the pure comparison group, the never-treated E
- “Stack” the cohort datasets and re-index time relative to each cohort's onset, so that the switches line up
- One saturated regression on the stack, with unit and period effects specific to each dataset; similar to interaction weighting, but with disproportionate weights across cohorts

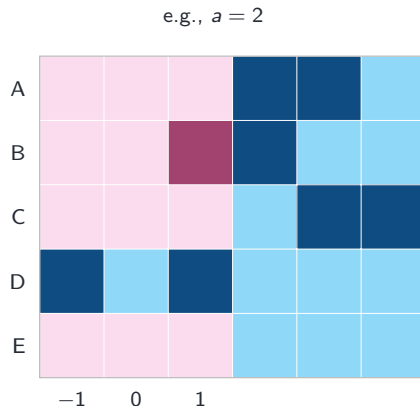


In practice a unit serves as a comparison for a cohort only while it stays untreated inside that cohort's event window (“clean controls”).

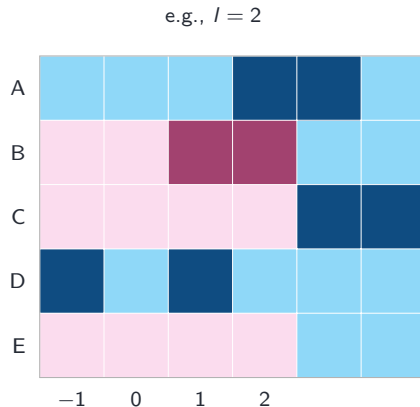
- No cohorts: one average effect **for switchers**, not the ATT; works with treatment reversal
- Switchers: (i, t) with $D_{it} \neq D_{i,t-1}$; here *B* and *D* at period 3
- Stable group: (i, t) with $D_{it} = D_{i,t-1}$, sharing the switchers' status in the previous period; here *A*, *C*, *E*
- DID_M: a DID from period 2 to 3, switchers against the stable group, estimates the **contemporaneous** effect at the period of the switch; leavers ($1 \rightarrow 0$) are compared with stable treated units in the same way, with the sign reversed
- Same data as PanelMatch on the next page: DID_M is PanelMatch with $a = l = 1$ and no refinement



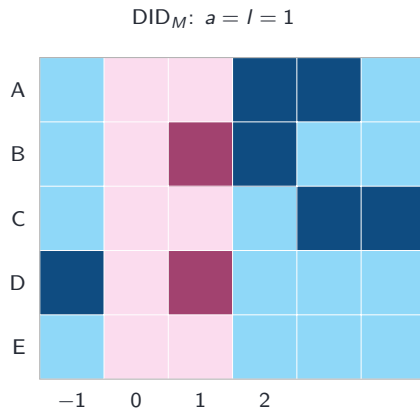
- General setting: match each treated (i, t) to units with the **same treatment history** over the previous a periods
- $a = 2$: for B at period 3 the matched set is $\{A, C, E\}$, untreated in periods 1–2 like B ; D (treated in period 1) is **excluded**
- A DID from period 2 to 3 within the matched set: the effect at $l = 1$
- Later periods $l = 2, 3, \dots$ (up to reversal): units that switch inside the window drop out; here A leaves at $l = 2$
- Refine the matched set on covariates X_{it}
- DID_M : the case $a = l = 1$ without refinement; at period 3 the switchers B and D are matched to A, C, E



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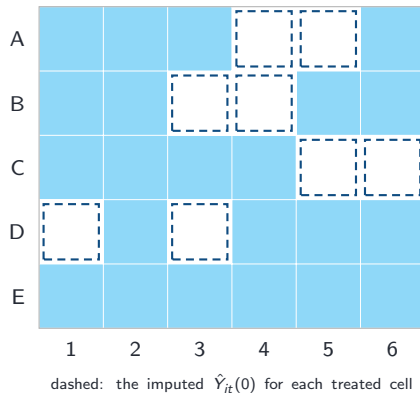


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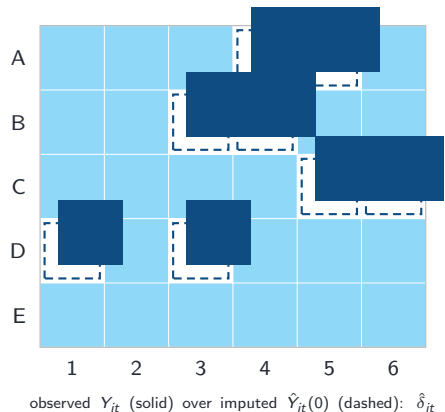
Imputation Methods (Borusyak et al. 2024; Liu, Wang, and Xu 2024)

- Also for the general setting, with a different logic: model the untreated outcome instead of choosing comparisons
- Set the treated cells aside and fit a model for $Y_{it}(0)$ on the **untreated cells only**; with two-way fixed effects, this is the fixed effects counterfactual estimator (FEct)
- Impute $\hat{Y}_{it}(0)$ for every treated cell
- One estimate per treated cell: $\hat{\delta}_{it} = Y_{it} - \hat{Y}_{it}(0)$
- Summarize the $\hat{\delta}_{it}$ however the question demands: overall ATT, dynamic effects, subgroups; no forbidden comparisons by construction; efficient under homoskedasticity (Borusyak et al. 2024)



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Extended TWFE, or the Two-Way Mundlak Regression (Wooldridge 2025)

- Keep the regression, but let the effect differ by cohort and period: one coefficient τ_{gt} for every cohort g in every post-treatment period $t \geq g$, instead of a single τ

$$Y_{it} = \alpha_i + \xi_t + \sum_g \sum_{t' \geq g} \tau_{gt'} \mathbf{1}\{G_i = g\} \mathbf{1}\{t = t'\} + \varepsilon_{it}$$

- “Mundlak”: in a balanced panel, replacing the unit effects by cohort dummies (the unit-level means of the regressors) gives numerically the same $\hat{\tau}_{gt}$, so the whole thing is one pooled regression
- Which cells compare: with no pre-period dummies, every pre-treatment cell stays in the untreated pool that pins down α_i and ξ_t , exactly as in imputation (IW and CSDID instead compare against one reference period); in a balanced panel without covariates the $\hat{\tau}_{gt}$ equal the imputation estimates
- Why it has caught on: one regression command with regression standard errors; covariates enter interacted with cohort and period; the same device extends to **nonlinear models** (Poisson for counts, logit for binary outcomes; Wooldridge 2023), where subtracting an imputed $\hat{Y}(0)$ makes no sense
- Limits: staggered adoption only (a cohort is a first-treatment date, so no reversal); many coefficients, to be aggregated as with CSDID; parallel trends still does the identifying work. Software: `etwfe` (R), `jwddid` (Stata)

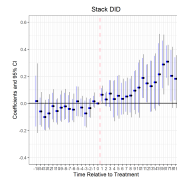
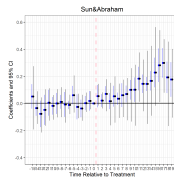
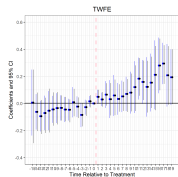
The Estimators, Side by Side

	DID extensions		Imputation methods	
	(2 × 2 DID as building blocks)		(outcome model w/ FE)	
Setting	Staggered	General	Staggered	General
Estimand	ATT	ATT for switchers	ATT by cohort-period	ATT
Estimator	IW, CSDID, stacked	PanelMatch, DID _M	extended TWFE (two-way Mundlak)	BJS, FE _{ct}
Comparisons	never/not-yet-treated	matched set	imputed counterfactual, one saturated regression	imputed counterfactual
Pre-treatment info	one reference period	the <i>a</i> previous periods	all untreated cells	all untreated cells
Key assumption	parallel trends	parallel trends	zero conditional mean or parallel trends	zero conditional mean or parallel trends

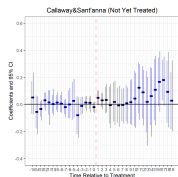
Which Estimator Should I Use?

- **Two-period or block assignment:** canonical DID is fine; TWFE equals DID here
- **Staggered, no reversal:** the full menu applies; IW/CSDID if cohort effects are the object, imputation for flexibility and diagnostics; extended TWFE when the outcome calls for a nonlinear model
- **Treatment reversal** (most political science panels): PanelMatch, DID_M , or imputation; the staggered-only tools do not apply
- **Few treated units or one treated unit:** move to synthetic control territory (Lecture 6)
- In well-powered data the choice rarely drives the answer; choose for **diagnostics and transparency**, then show robustness across estimators

One Staggered Application, Five Estimators

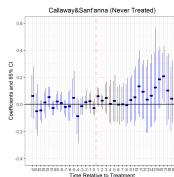


TWFE event study



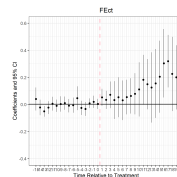
CSDID (not-yet-treated)

Sun and Abraham



CSDID (never-treated)

stacked DID

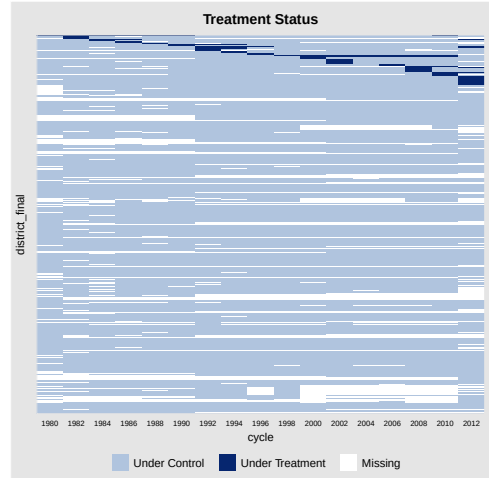


imputation (FEct)

- Bischof and Wagner (2019): the estimators tell one story; when parallel trends looks plausible, agreement is the norm

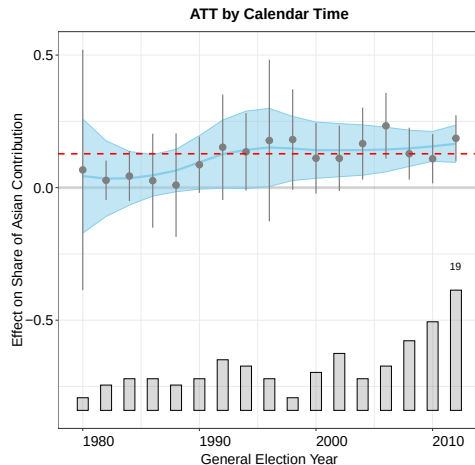
Example 1: Coethnic Mobilization (Grumbach and Sahn 2020)

- Do ethnic minority candidates mobilize coethnic donors?
 - Unit of analysis: 498 US congressional districts, 1980–2012
 - Treatment: Asian (vs. White, Black, and Hispanic) candidates in general elections
 - Outcome: percentage of Asian donations
- The treatment matrix is unbalanced with missing data; the imputation approach handles it without reweighting gymnastics

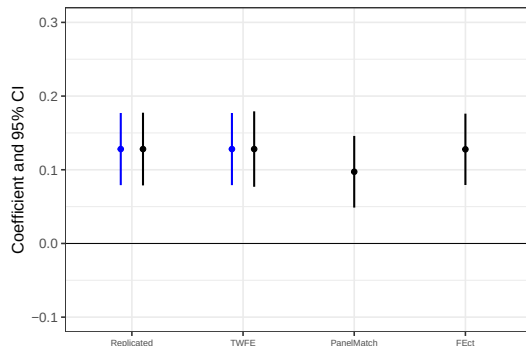


Effects Are Heterogeneous

- Effects grow over calendar time: exactly the heterogeneity that breaks the single- τ TWFE regression
- Do the HTE-robust estimators change the answer here?



Do the Estimators Agree in Practice?

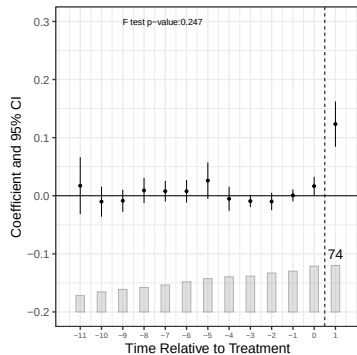


- Grumbach and Sahn (2020): TWFE, PanelMatch, and FEct give similar answers; here the design is strong and HTE is second-order

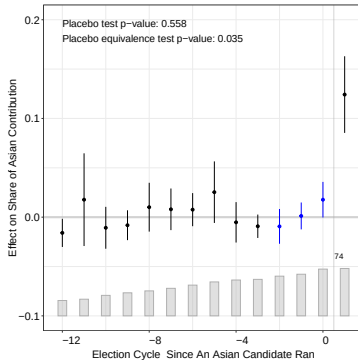
The imputation approach turns identification checks into estimable quantities:

- **Pre-trend F test:** are the average $\hat{\delta}_{it}$ in pre-treatment periods jointly zero?
- **Placebo test:** hide the last few pre-treatment periods from estimation, then check the “effect” in the hidden periods
- **Carryover test:** hide the first few **post-exit** periods, then check for lingering effects after treatment ends
- **Equivalence (TOST) tests:** instead of failing to reject zero, reject pre-trends larger than a tolerance; more informative in well-powered settings

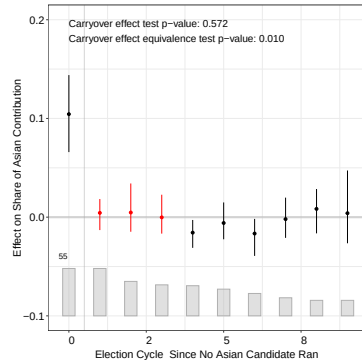
The Suite Applied to Example 1 (Grumbach and Sahn 2020)



F test for no pre-trend



placebo test



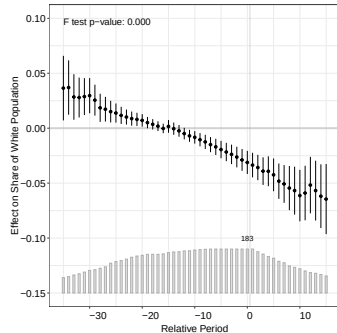
carryover test

- None of the three rejects its null here; with 74 treated observations, power is the caveat to keep in mind

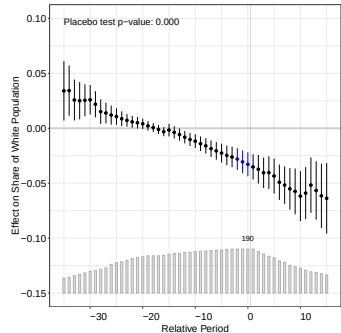
Example 2: Lawsuits against Land-Use Restrictions (APSR 2020)

- Treatment: a Fair Housing Act lawsuit against a city's land-use restrictions; outcome: the share of the city's residents who are non-Hispanic white (California cities, 1970–2010, staggered)
- The imputation event-study plot slopes downward **before** treatment: the pre-trend F test and the placebo test both reject ($p < 0.001$)

dynamic effects, with the pre-trend F test

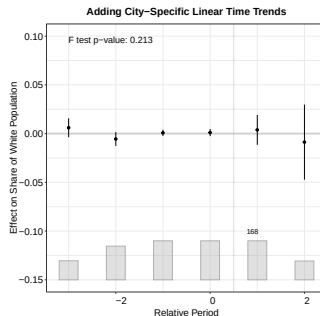
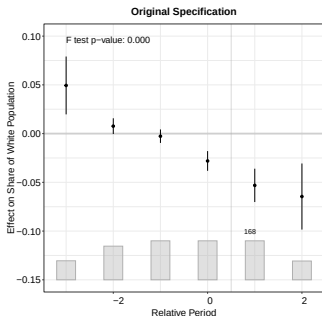


placebo test (blue: the hidden periods)

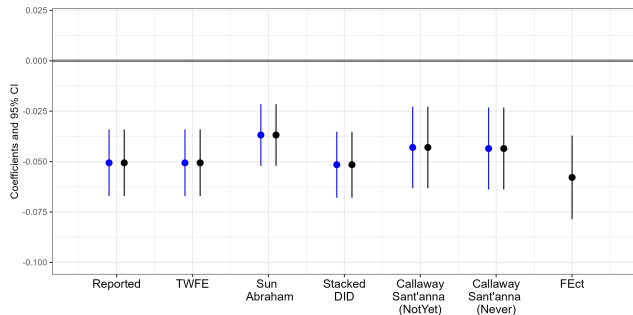


Example 2: Adding City-Specific Linear Time Trends

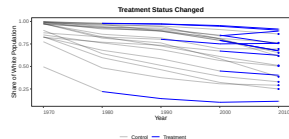
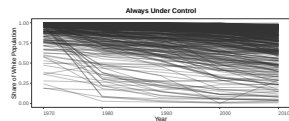
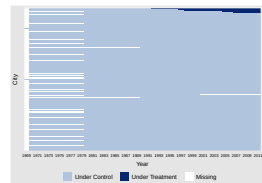
- The demographic series are mostly interpolated between Census years; dropping the interpolated years changes little
- Adding **city-specific linear time trends** removes the pre-trend (F test $p = 0.21$) and the negative effect with it, on the same -3 to $+2$ window
- The downward slope predates the lawsuits; once the trends absorb it, nothing is left



Example 2: The Estimators Still Broadly Agree

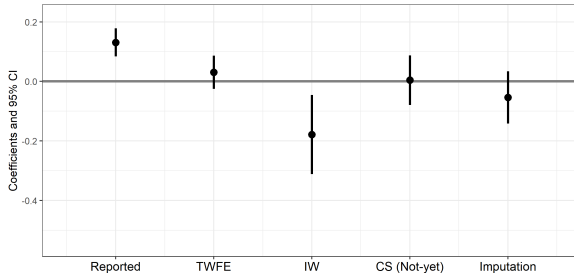


- Seven estimators, one answer, about -0.05 : they all inherit the **same** violated assumption
- Agreement across estimators is not robustness; the raw-data plots and the two tests catch what comparing estimators cannot

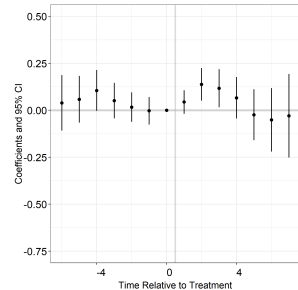


Example 3: Updating Cadastral Maps and Property Tax Revenue (JOP 2021)

- The authors ask whether updating the cadastral map, the register of properties, raises property tax revenue in Brazilian municipalities (2004–2015, staggered adoption)
- Full-sample reanalysis: the estimators **disagree**; the reported estimate is positive, TWFE and CSDID sit near zero, interaction weighting is negative
- The paper's event-study specification, replicated on its subsample, shows the positive effect

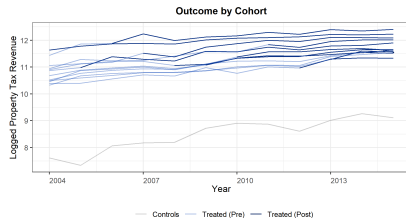
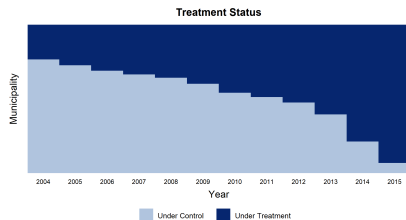


replicated on the full sample

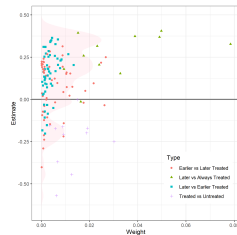


the paper's specification on its subsample

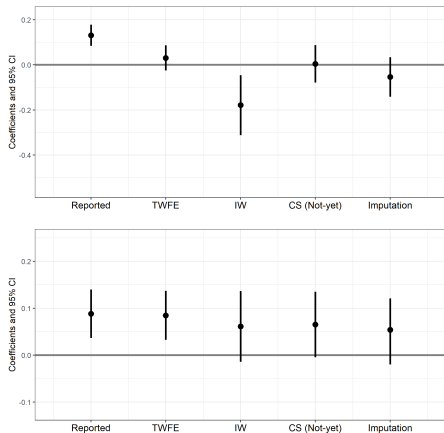
Example 3: Look Deeper



- A large **always-treated** block sits on top of the staggered adopters; only a handful of municipalities are never treated
- Those never-treated municipalities collect far less revenue than every treated cohort: a weak comparison group
- Goodman-Bacon's decomposition of the TWFE estimate: comparisons among treated cohorts sit around $+0.25$, comparisons against the never-treated around -0.4 ; the answer depends on **which comparisons carry weight**

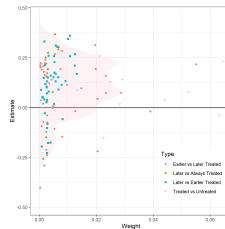


Example 3: A Trimmed Subsample



top: full sample; bottom: trimmed subsample

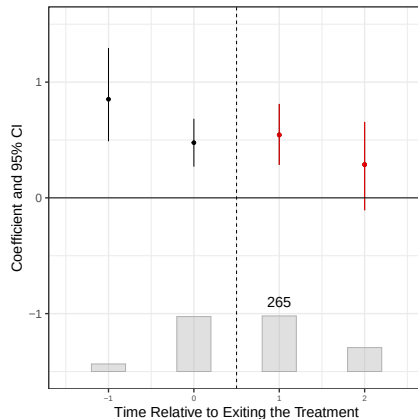
- Drop the always-treated block and the far-away never-treated units, without looking at the outcome
- On the subsample the estimators **converge**, all between 0.05 and 0.09 with overlapping intervals; the comparisons now form one cloud
- The disagreement was a symptom of a weak comparison set, not of heterogeneity; trimming to “compare like with like” is design work (Imbens and Rubin 2015)



- **Example 1, coethnic mobilization:** strong design; heterogeneity matters marginally; the estimators, TWFE included, broadly agree
- **Example 2, land-use lawsuits:** clear signs of a parallel trends violation; heterogeneity is second-order; agreement among estimators does not mean robustness; plotting the raw data and the two tests spot the problem
- **Example 3, cadastral maps:** when estimators disagree, suspect the comparison set before the estimators; design-phase trimming, done without the outcome, restores agreement

Carryover Effects Are Common

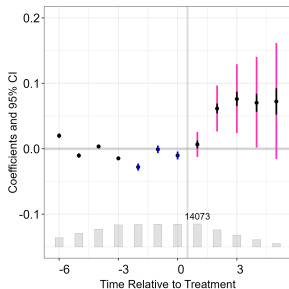
- With treatment reversal, effects that linger after exit violate the no-carryover assumption; we find this in roughly a third of the reanalyzed studies
- When carryover is plausible, estimate effects relative to treatment **entry** only, or model the exit explicitly



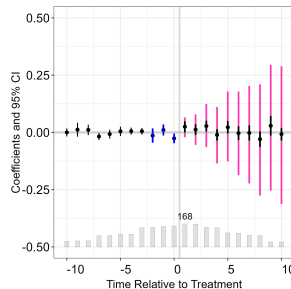
Distelhorst and Locke (2018): rejects no carryover — unlike Example 1
(Grumbach and Sahn 2020)

Sensitivity Analysis: Relaxing Parallel Trends

- Rambachan and Roth (2023): allow post-treatment confounding up to M times the largest change between neighboring placebo periods; $M = 0$ means parallel trends holds exactly



Hall and Yoder (2022)

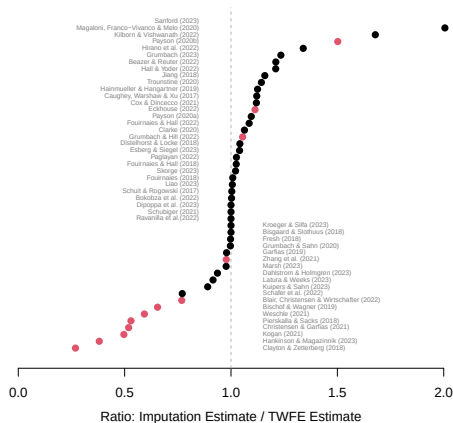


Caughey, Warshaw, and Xu (2017)

- The question shifts from “does the effect exist” to “how much confounding would it take to erase it”

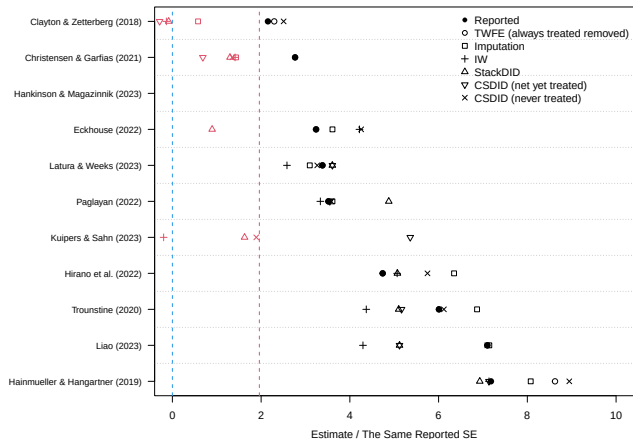
- We replicated a main result from top political science journals, 2017–2023: 102 papers use linear panel models; 64 have a “proper” TWFE design; 49 are replicable
- Each study re-estimated with the full menu: IW, CSDID, stacked, PanelMatch/DID_M, and imputation, plus the diagnostic suite and sensitivity analysis
- Settings in the sample: 55% general (with reversal), 27% staggered, 12% multi-period block, 6% classic 2×2 : the staggered-only estimators do not even apply to most studies

Finding 1: Estimates Rarely Flip, but Variability Exists



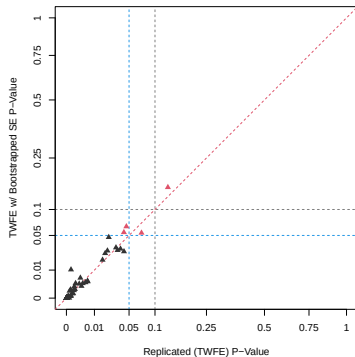
- The ratio of imputation to TWFE estimates centers on 1.0 (mean 1.02, median 1.00); large deviations concentrate in sparse-data cases

Finding 1 (continued): The Staggered Cases

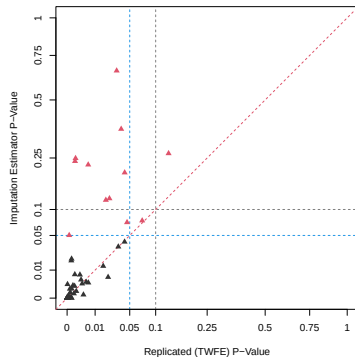


- Across the staggered studies, the five HTE-robust estimators and TWFE broadly agree on magnitudes

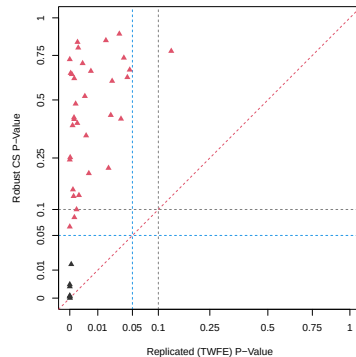
Finding 2: “Robust” DID Is Power Hungry



bootstrapped SEs (91% survive)



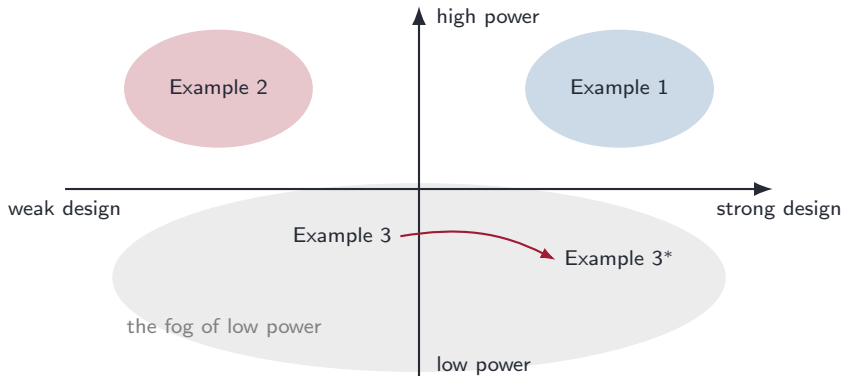
+ relaxing constant effects (75%)



+ relaxing parallel trends (23%)

- Each layer of robustness costs power; fewer than a third of the findings survive mild sensitivity analysis

Big Picture: Design and Power



- Estimator choice barely moves a study along either axis; **design** and **power** do. Trimming moved Example 3 to the right; moving up takes more data

The Paradox of Committing to TWFE

- Clear signs of parallel trends violations in roughly a quarter of the studies; in over half, we **cannot tell**, because there are too few pre-periods or too little power
- With enough power, you can afford an HTE-robust estimator, so use one
- Without enough power, you cannot **validate** the assumptions TWFE needs anyway
- Either way, blind TWFE is hard to defend; the binding constraints are **design and power**, not the estimator

Recommendations

	Do	Don't
Design	Start from a research design; check for feedback from past outcomes	Start by running regressions on whatever data exist
Raw data	Plot treatment status and outcome trajectories first	Run regressions without looking at the data
Estimators	Use HTE-robust estimators; always plot dynamic effects	Choose models on belief; report coefficients only
Diagnostics	Run visual and statistical tests of the assumptions	Skip visualization and diagnostics
Clustering	Cluster at the level of treatment assignment or higher	Cluster below the assignment level
Small clusters	Cluster-bootstrap when clusters $\lesssim 50$	Rely on asymptotic SEs

- The HTE-robust menu is organized by comparisons: DID extensions choose their 2×2 s explicitly; imputation methods model $Y(0)$ from untreated cells
- In 49 reanalyses, estimator choice rarely flips results; parallel trends violations and **low power** are the first-order problems
- Run the diagnostic suite (pre-trend, placebo, carryover, equivalence) and report a sensitivity analysis, not just a point estimate

- `fect` (R & Stata): imputation estimators with the full diagnostic suite
- `panelView` (R & Stata): treatment status and outcome visualization
- `did`, `PanelMatch`, `DIDmultiplegt`: the DID extensions; `didimputation`: BJS imputation; `etwfe` (R) and `jwddid` (Stata): extended TWFE
- Tutorials: <https://yiqingxu.org/tutorials/panel.html>

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